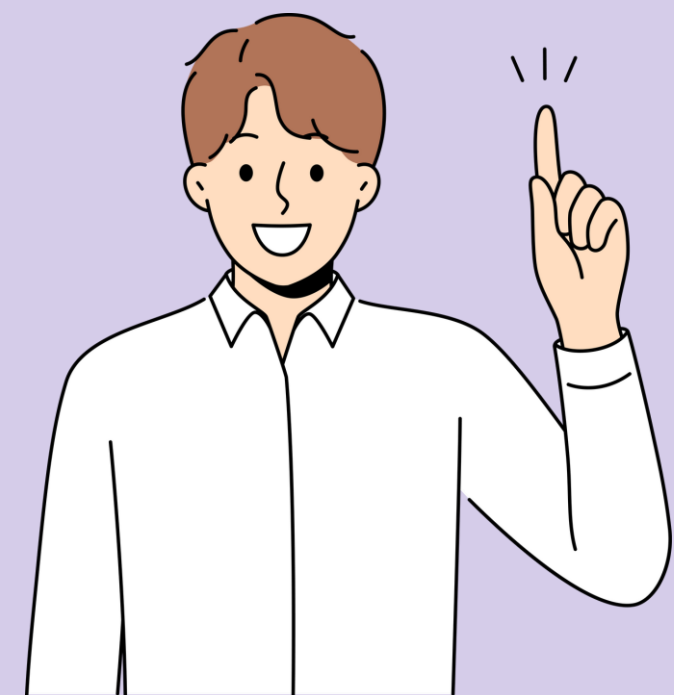


LECTURE 2

LINEAR REGRESSION

= YOUR FIRST ML MODEL

Predictive Market Modeling



INTRODUCTION

Consider this general scenario

House Size (sq ft)	Number of Rooms	Number of Floors	Distance to City Center (km)	House Price (\$)
1000	3	1	10	150,000
1500	4	2	8	250,000
800	2	1	15	120,000
2000	5	2	5	400,000
1200	3	1	12	180,000
1700	4	2	7	300,000

Can we predict house prices for new homes based on size, rooms, floors, and distance?

Well.... Isn't it obvious :)

APPROCHING THE HOUSING PROBLEM

BASIC FLOWCHART

Define the Problem
Collect and Analyze Data
Choose a Model
Estimate Parameters
Define the Cost Function
Optimize the Model
Evaluate the Model
Make Predictions

Defining The Problem

Objective: Predict house prices based on specific features.

Identify the dependent variable (target): House Price.

Identify the independent variables (features):

House Size (sq ft)

Number of Rooms

Number of Floors

Distance to City Center

Analyzing DaTa

Data Cleaning

- Handle missing values by imputing or removing rows/columns.
- Remove outliers if they are errors or irrelevant to the problem.

Data Transformation

- Normalize or standardize numerical features to bring them to the same scale.
- Encode categorical variables (if any) using techniques like one-hot encoding.

Exploratory Data Analysis (EDA)

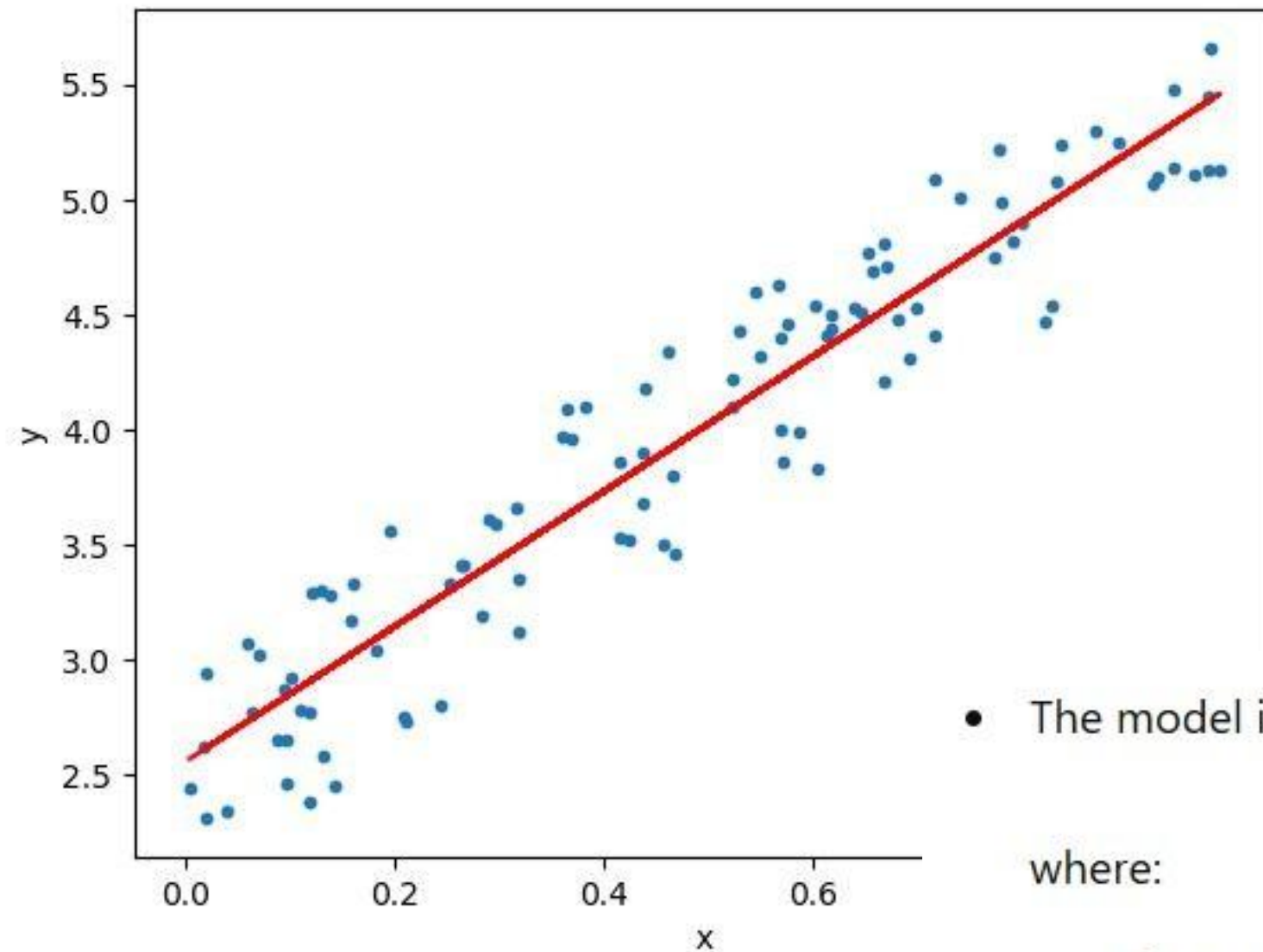
- Use summary statistics (mean, median, variance) to understand data distribution.
- Visualize relationships using scatter plots (e.g., size vs. price).
- Check for correlations between features and the target variable.

Splitting the Data

- Divide the dataset into training and testing subsets (e.g., 80%-20%).
- Use training data for model building and testing data for evaluation.

Model

A model in machine learning is a mathematical representation of a system that describes the relationship between input (features) and output (target).



- The model is typically expressed as:

$$f(X; \theta)$$

where:

- f : The functional form of the model (e.g., linear regression, neural network, etc.).
- θ : The parameters of the model (e.g., coefficients in linear regression or weights in neural networks).

OUR MODEL (CRLM)

The Linear Model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \epsilon_i$$

Assumptions

1. **Linearity:** $y = W^T X + B + \epsilon$, where W (weights) and B (bias) are the parameters to estimate.
2. **Exogeneity:** $E[\epsilon|X] = 0$, meaning errors are uncorrelated with predictors.
3. **Homoscedasticity:** $\text{Var}(\epsilon) = \sigma^2$, constant error variance.
4. **No autocorrelation:** Errors are uncorrelated across observations.
5. **Normality:** $\epsilon \sim N(0, \sigma^2)$.

ESTIMATION OF PARAMETERS

$$y = W^T X + B$$

Here W (weights) and B (bias) are the parameters to be estimated

OLS Estimator

Minimize the residual sum of squares (RSS)

$$RSS = \sum_{i=1}^n (y_i - W^T X_i - B)^2$$

Derivation

1. Define the cost function:

$$J(W, B) = \frac{1}{2} \sum_{i=1}^n (y_i - W^T X_i - B)^2$$

2. Differentiate J w.r.t. W and B :

$$\frac{\partial J}{\partial W} = -X^T (y - XW - B)$$

$$\frac{\partial J}{\partial B} = -\sum_{i=1}^n (y_i - W^T X_i - B)$$

3. Solve for optimal W and B :

$$W = (X^T X)^{-1} X^T y$$

$$B = \bar{y} - W^T \bar{X}$$

Maximum Likelihood Estimation (MLE)

Key Idea:

- Likelihood function:

$$L(\beta) = \prod_{i=1}^n P(y_i | X_i, \beta)$$

- Log-likelihood:

$$\ln L(\beta) = \sum_{i=1}^n \ln P(y_i | X_i, \beta)$$

- MLE finds β that maximizes $\ln L(\beta)$.

$$\epsilon \sim N(0, \sigma^2) \implies Y_i \sim N(W^T X_i + B, \sigma^2)$$

$$L(W, B, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - W^T X_i - B)^2}{2\sigma^2}\right)$$

Log-Likelihood Function:

$$\ell(W, B, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - W^T X_i - B)^2$$

Hypothesis Testing

t-Test

- Purpose: Test significance of individual coefficients.
- Null Hypothesis: $H_0 : W_j = 0$.
- Test Statistic:

$$t = \frac{\hat{W}_j}{SE(\hat{W}_j)}, \quad t \sim t_{n-k-1}$$

- Example: If t -statistic $>$ critical t -value, reject H_0 .

Hypothesis Testing

z-Test

- Purpose: Test large-sample significance of coefficients.
- Null Hypothesis: $H_0 : W_j = 0$.
- Test Statistic:

$$z = \frac{\hat{W}_j}{SE(\hat{W}_j)}, \quad z \sim N(0, 1)$$

- Example: For $\alpha = 0.05$, if $|z| > 1.96$, reject H_0 .

Hypothesis Testing

F-Test

- Purpose: Test overall model significance.
- Null Hypothesis: $H_0 : W_1 = W_2 = \dots = W_k = 0$.
- Test Statistic:

$$F = \frac{(RSS_0 - RSS_1)/k}{RSS_1/(n - k - 1)}, \quad F \sim F_{k, n-k-1}$$

- Example: If $F > F_{\text{critical}}$, reject H_0 .

THANK YOU :)